SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1021	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	S.Venkateswarlu	**Reg. No.:**	192425340
Date:	14/10/2026	**Duration:**	60 minutes

Code of Conduct

I S.Venkateswarlu_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *S.Venkateswarlu_____*

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. **Identify the Project Vision and Stakeholders**
   - Define the project's vision, objectives, and identify all key stakeholders involved.
2. **Create Epics and User Stories**
   - Break down the requirements into Epics and formulate well-structured User Stories using the format: *As a <user>, I want <feature>, so that <benefit>.*
3. **Define Acceptance Criteria**
   - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
4. **Prioritize the Product Backlog**
   - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
5. **Plan the Sprint Backlog**
   - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
6. **Estimate Effort Using Story Points**
   - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
7. **Present the Sprint Goal and Expected Deliverables**
   - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification **(30%)**	/5
Epics & User Stories **(25%)**	/5
Acceptance Criteria **(20%)**	/5
Product Backlog Prioritization & Sprint Planning **(15%)**	/5
Story Point Estimation, Sprint Goal & Presentation **(10%)**	/5
TOTAL	/25

Industry Context: International seaports are highly complex bottlenecks where giant container ships must be loaded and unloaded quickly. Inefficient container yard management leads to crane delays, traffic jams for terminal trucks, and long ship wait times, costing shipping companies thousands of dollars per hour. Port operators need a centralized, automated orchestration system that schedules crane movements and container placements to maximize port speed.

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Problem Statement: Design and develop a highly responsive Seaport Container Yard Coordination Platform that orchestrates container movements, crane schedules, and cargo truck traffic. The system must process real-time inputs—including ship cargo configurations, automated guided vehicle (AGV) paths, container yard storage grids, and import/export timelines. It must compute optimal crane picking patterns to minimize ship unloading times.

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Objectives:

o
Balance
container placements inside yard storage grids to minimize redundant movements.

o
Coordinate
crane movements in real time to avoid collisions and mechanical downtime.

o
Route
automated terminal trucks through cargo yards using dynamic traffic paths.

o
Adapt
yard logistics automatically when ship arrivals shift due to weather delays.

o
Provide
live operational dashboards for port logistics and customs management teams.

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Expected Outcomes:

o
Shorter
turnaround times for cargo ships at international maritime shipping ports.

o
Maximum
performance and utilization from high-value port cranes and automated trucks.

o
Lower
traffic gridlock and fuel waste for terminal transport vehicles inside ports.

o
Increased
cargo processing capacities for ports without expanding physical footprints.

1. INTRODUCTION AND INDUSTRY CONTEXT

A container terminal is a connected logistics environment. Containers move from ship to quay crane, then by truck or AGV to a yard block, and later to a gate, rail connection or another vessel. Export containers follow a related path in the opposite direction. Each move depends on space, equipment availability, cargo priority, safety rules and a time window.

1.1 Industry context and rationale

- Global maritime transport depends on predictable port calls and coordinated cargo transfers.
- Terminal equipment and land are costly resources, so idle time and avoidable re-handling matter.
- Weather, late vessels, equipment faults and traffic incidents can invalidate a plan quickly.
- Digital coordination supports visibility, traceability and evidence-based operational decisions.

1.2 Project boundary

The initial deliverable is a software prototype and simulation, not a certified control system for directly operating cranes or vehicles. Live equipment integration requires port-specific interfaces, cybersecurity review, safety engineering, testing and formal operational approval.

2. PROJECT VISION, OBJECTIVES AND SCOPE

2.1 Project vision

To provide port teams with one reliable operational view and a decision-support platform that coordinates container placement, crane schedules and terminal-vehicle routes, helping move cargo safely and efficiently despite changing conditions.

2.2 Business objectives

ID	Objective	How success can be measured
O1	Reduce avoidable container re-handling.	Compare re-handling moves per container against a simulated baseline.
O2	Improve crane coordination.	Measure crane idle time, conflicts detected and schedule adherence.
O3	Reduce vehicle congestion.	Measure average route delay, queue length and blocked-lane events.

O4	React to disruptions.	Measure time from a disruption input to a revised feasible plan.
O5	Improve operational visibility.	Measure dashboard freshness, data completeness and alert acknowledgement.
O6	Maintain safe human oversight.	Record overrides, access permissions and safety-rule violations; target zero unsafe recommendations in tests.

2.3 In scope

- Vessel schedule and cargo-manifest entry/import.
- Yard-grid representation with occupancy and container attributes.
- Basic container-to-slot recommendations based on constraints and priorities.
- Crane work queue and conflict alerts in a simulated environment.
- AGV/terminal-truck route suggestions and lane status.
- Weather/late-arrival disruption entry and plan recalculation.
- Role-based dashboards, audit logs, reports and notifications.

2.4 Out of scope for the first release

- Unsupervised control of real cranes, trucks or AGVs.
- Replacement of customs, billing, shipping-line or terminal operating systems.
- Guaranteed global optimum for every operational scenario.
- Automatic legal or safety decisions without a responsible human authority.

3. KEY STAKEHOLDERS AND RESPONSIBILITIES

Stakeholders are the people, teams and external organizations that use, supply data to, approve or are affected by the platform. Their needs can conflict; for example, vessel planners may prioritize rapid unloading while yard planners need storage stability.

Stakeholder	Main responsibility	Platform need / concern
Port operations manager	Own terminal-level performance and policies.	Throughput, visibility, service-level metrics and escalation.

Vessel/berth planner	Manage vessel calls and berth windows.	Accurate ETA, cargo priorities and feasible work sequence.
Quay-crane supervisor	Allocate crane work and maintain safe separation.	Conflict-free schedules, equipment status and manual override.
Yard planner	Assign storage blocks and retrieval priorities.	Slot availability, container compatibility and fewer reshuffles.

3.1 Stakeholder engagement

- Conduct short workshops with planners and equipment supervisors before finalizing workflows.
- Use prototype reviews to validate screen layouts and operational terminology.
- Agree on data ownership, update frequency and responsibility for correcting inaccurate records.
- Review safety-critical decisions with qualified port personnel before any operational trial.
- Provide training and a feedback channel for users during pilot testing.

3.2 Main stakeholder challenge

The major challenge is shared responsibility across departments. The system should not assume that one team has complete or always-current information. It should show data timestamps, identify missing fields, flag contradictory inputs and make ownership of corrections clear.

4. REQUIREMENTS AND EPICS

An epic is a large capability that can be divided into smaller user stories. User stories describe value from a user's perspective. The epics below organize the product around operational outcomes rather than technical components alone.

Epic ID	Epic	Purpose	Priority
E1	Vessel and cargo intake	Register vessel calls, cargo manifests, ETAs and deadlines.	Must
E2	Yard visibility and slotting	Show storage blocks and recommend compatible empty slots.	Must

E3	Crane scheduling	Build work queues and identify time/space conflicts.	Must
E4	Vehicle routing	Suggest feasible paths and avoid blocked/congested lanes.	Should
E5	Disruption management	Recalculate plans after delays, breakdowns or closures.	Should

4.1 Epic dependencies

Vessel and cargo data (E1) and yard-state data (E2) are prerequisites for meaningful crane and route recommendations. Security (E7) must be built into the first increment rather than added at the end. Dashboard work (E6) can start early with sample data, then connect to validated services as they become available.

4.2 Functional requirements

- FR1: Authorized users can create, view and update vessel calls and cargo records.
- FR2: The system validates required cargo identifiers and flags duplicate or incomplete records.
- FR3: Yard blocks show occupied, empty, reserved and unavailable slots.
- FR4: The system proposes container slots that satisfy configured constraints.
- FR5: Crane plans identify job order, time window, equipment availability and conflicts.
- FR6: Route planning avoids blocked lanes and respects configured vehicle restrictions.
- FR7: Authorized users can record disruptions and request a revised plan.
- FR8: Dashboards show data freshness, current status, alerts and key indicators.
- FR9: Significant plan changes and manual overrides are audit logged.

5. USER STORIES — VESSEL, YARD AND CRANE

The standard user-story format is: “As a [user], I want [feature], so that [benefit].” The stories below are deliberately small enough to estimate and test.

5.1 Epic E1 — Vessel and cargo intake

ID	User story	Value
US-01	As a vessel planner, I want to register a vessel call with ETA and berth window, so that the terminal can prepare resources.	Earlier preparation
US-02	As a cargo clerk, I want to upload or enter a cargo manifest, so that container work can be planned from a shared record.	Fewer manual hand-offs
US-03	As a planner, I want incomplete or duplicate cargo records flagged, so that incorrect data does not enter the plan.	Data quality

5.2 Epic E2 — Yard visibility and slotting

ID	User story	Value
US-04	As a yard planner, I want to view the yard grid and slot status, so that I can identify available storage space.	Shared yard visibility
US-05	As a yard planner, I want compatible empty slots recommended for a container, so that future retrieval requires fewer extra moves.	Less re-handling
US-06	As an operator, I want to mark a slot unavailable, so that damaged or restricted areas are not assigned.	Constraint compliance

5.3 Epic E3 — Crane scheduling

ID	User story	Value
US-07	As a crane supervisor, I want a prioritized crane work queue, so that urgent cargo jobs are visible.	Better sequencing
US-08	As a crane supervisor, I want potential schedule and separation conflicts flagged, so that I can review them before work is assigned.	Safer planning
US-09	As an operations manager, I want to compare planned and actual job times, so that bottlenecks can be identified.	Performance insight

6. USER STORIES — ROUTING, DISRUPTIONS AND DASHBOARDS

6.1 Epic E4 — Vehicle routing

ID	User story	Value
US-10	As a dispatcher, I want a suggested route for each vehicle job, so that containers reach their destination efficiently.	Reduced travel
US-11	As a dispatcher, I want blocked or congested lanes excluded from new routes, so that vehicles avoid known obstructions.	Fewer delays
US-12	As a supervisor, I want route changes explained, so that operators understand why a path was updated.	Trust and clarity

6.2 Epic E5 — Disruption management

ID	User story	Value
US-13	As a vessel planner, I want to update a delayed ETA, so that affected work can be rescheduled.	Adaptability
US-14	As a dispatcher, I want to mark a lane or vehicle unavailable, so that new assignments use feasible resources.	Continuity
US-15	As an operations manager, I want a revised plan and a summary of affected jobs, so that I can approve recovery actions.	Faster recovery

6.3 Epic E6 — Dashboard and alerts

ID	User story	Value
US-16	As an operations manager, I want a live dashboard of vessels, cranes, yard occupancy and vehicles, so that I can see terminal status in one place.	Situational awareness
US-17	As a supervisor, I want alerts for stale data and operational conflicts, so that I can investigate exceptions quickly.	Early warning
US-18	As a team member, I want to acknowledge and comment on an alert, so that the response is traceable.	Accountability

6.4 Epic E7/E8 — Security and analytics

ID	User story	Value
US-19	As an administrator, I want role-based access, so that users can only perform permitted actions.	Protection
US-20	As an auditor, I want plan changes and overrides recorded with user and timestamp, so that decisions can be traced.	Accountability
US-21	As an operations manager, I want reports comparing baseline and proposed plans, so that improvements can be evaluated.	Evidence-based decisions

7. ACCEPTANCE CRITERIA — METHOD AND SELECTED STORIES

Acceptance criteria are specific conditions that must be met before a story is accepted. They make requirements testable and reduce disagreements between users, developers and testers. Given–When–Then describes the starting context, action and expected result.

7.1 US-01: Register a vessel call

Given an authenticated vessel planner and valid vessel details, when the planner saves a new call with a unique vessel/call identifier, ETA and berth window, then the system stores the call and displays it in the vessel list. If required fields are missing or the identifier is duplicated, the system rejects the save and explains the validation error.

7.2 US-03: Validate cargo records

Given a manifest containing container identifiers and required attributes, when the user imports it, then the system reports accepted rows, rejected rows and reasons. Duplicate identifiers and missing mandatory fields must be flagged. The import must not silently discard rejected rows.

7.3 US-04: Display yard grid

Given a yard block with configured slots, when the user opens the yard view, then every configured slot is shown with a clear state: empty, occupied, reserved or unavailable. The displayed update time must be visible. The view must not label unknown status as empty.

7.4 US-05: Recommend a compatible slot

Given a container and current yard state, when the planner requests a recommendation, then the system returns only slots satisfying configured size, weight, hazard, reefer and availability constraints. If no feasible slot exists, it returns a clear no-feasible-slot message instead of an invalid assignment.

7.5 US-07: Prioritized crane queue

Given jobs with priority, deadline, crane availability and safety constraints, when the supervisor generates a queue, then jobs are ordered by the configured policy and each item shows its reason/priority. Jobs that cannot be scheduled are listed separately with a reason.

8. ACCEPTANCE CRITERIA — ROUTING, DISRUPTIONS AND DASHBOARDS

8.1 US-08: Detect crane conflicts

Given a proposed crane schedule, when the schedule validator runs, then it flags overlapping assignments or violations of configured separation and resource constraints. A plan with a detected safety-critical conflict cannot be marked approved until corrected or explicitly handled through the port's authorized safety process.

8.2 US-10/US-11: Vehicle route suggestion

Given a vehicle origin, destination, vehicle type and current lane status, when a route is requested, then the system returns a connected path using permitted lanes. Known blocked lanes must not appear in a newly generated route. If no route is feasible, the system reports that condition and does not fabricate a path.

8.3 US-13: Update delayed ETA

Given a scheduled vessel call and dependent work, when an authorized planner updates the ETA, then the system identifies affected jobs and marks the existing plan as needing review. The system offers a revised schedule for approval and preserves the previous plan in the audit history.

8.4 US-16: Live dashboard

Given available vessel, crane, yard and vehicle data, when an authorized user opens the dashboard, then the system displays the latest received values, their timestamps and the current exception count. A configurable freshness target—for example, a 10-second refresh in the prototype—must be tested and reported; this is a proposed target, not a universal port standard.

9. PRODUCT BACKLOG PRIORITIZATION

The Product Backlog is the ordered list of work that may be needed in the product. Prioritization considers business value, risk reduction, dependencies and user need. The MoSCoW method groups items as Must, Should, Could or Won't for the current release. Priority is revisited as evidence and stakeholder feedback arrive.

Rank	Story	Priority	Reason / dependency	Estimate
1	US-19 Role-based access	Must	Security foundation for all other features.	5
2	US-01 Register vessel call	Must	Core vessel data for planning.	3
3	US-02 Enter/import cargo manifest	Must	Cargo jobs depend on reliable cargo data.	5
4	US-03 Validate cargo data	Must	Prevents bad data spreading into schedules.	3
5	US-04 Display yard grid	Must	Required for yard visibility and slotting.	5

9.1 Prioritization rules

- Deliver security, core data and visibility before advanced optimization.
- Prioritize work that removes dependencies for several later stories.
- Do not treat story points as business value; they estimate effort and uncertainty.
- Keep safety and data-integrity requirements mandatory even when they do not create visible screens.
- Use simulation evidence and stakeholder feedback to reorder the backlog.

10. COMPARISON TABLE: CURRENT APPROACH VS PROPOSED PLATFORM

Dimension	Disconnected/manual approach	Proposed coordination platform	Expected benefit
Operational view	Separate spreadsheets, messages or subsystem screens.	Shared dashboard with timestamped status.	Fewer information gaps.
Yard placement	May rely on local judgement and manual checking.	Constraint-based slot recommendation.	Potentially fewer reshuffles.

Crane planning	Schedules may be updated separately.	Common job queue and conflict checks.	Better coordination and visibility.
Vehicle movement	Routes may be assigned without current lane context.	Route suggestions use known closures and congestion state.	Less avoidable route delay.
Disruption response	Teams communicate changes and rebuild plans manually.	Impact list and revised plan for review.	Faster, traceable recovery.

11. SYSTEM ARCHITECTURE AND DATA FLOW

11.1 Logical architecture

Layer	Components	Responsibility
User interface	Operations dashboard, planner screens, alert panel	Display state and collect authorized actions.
API and identity	API gateway, authentication, role checks	Protect endpoints and validate requests.
Application services	Vessel/cargo, yard, crane, route, disruption services	Apply business rules and coordinate features.
Planning/optimization	Constraint checker, heuristics, scheduling engine	Generate candidate plans and explain infeasibility.
Data layer	Operational database, event/audit store	Persist current state, history and changes.
Integration layer	Adapters for terminal systems, sensors and feeds	Normalize external data; manage failures and stale inputs.
Monitoring	Logs, metrics, health checks and alerts	Observe reliability, performance and errors.

11.2 High-level data flow

External systems and authorized users provide vessel, cargo, yard, equipment and lane updates. An ingestion layer validates the schema and timestamps. Valid events update the operational state and event history. Planning services use this state to generate candidate schedules. A constraint validator checks the

candidate against configured rules. The dashboard presents the plan, conflicts and explanations. Only authorized actions publish or override plans.

11.3 Data entities

Entity	Example fields	Purpose
VesselCall	call_id, vessel_id, ETA, berth_window, status	Track a port call.
Container	container_id, size, weight, type, hazard/reefer flags, destination	Describe cargo constraints.
YardSlot	block, row, bay, tier, status, restrictions	Represent available storage.
Equipment	equipment_id, type, status, capacity, location	Track cranes and vehicles.

12. NON-FUNCTIONAL REQUIREMENTS AND QUALITY

Requirement	Proposed target / test	Reason
Responsiveness	Prototype dashboard refresh target: 10 seconds or less under defined test load.	Keep information useful for supervisors.
Availability	Define a pilot target with the port IT team; monitor uptime and failed requests.	Operational users need dependable access.
Security	Authentication, role-based authorization, encrypted transport and secrets management.	Protect cargo and operational data.
Integrity	Validate identifiers, units, timestamps and required fields; reject unsafe defaults.	Bad data can create invalid plans.
Auditability	Record critical changes and overrides with user and timestamp.	Support incident review.

13. SPRINT PLANNING: FIRST SPRINT

A Sprint is a fixed-length development period. For this proposal, Sprint 1 is assumed to last two weeks. The sprint is a planning example; the team should adjust scope based on its capacity, skills and actual project constraints.

13.1 Sprint objective

Create a secure and demonstrable minimum foundation that allows an authorized planner to register a vessel call, record basic cargo data, view a simple yard grid and see the combined state on a dashboard.

13.2 Selected sprint stories

Story	Scope in Sprint 1	Points	Reason selected
US-19	Basic role-based access for planner/admin roles.	5	Security foundation.
US-01	Create and list vessel calls with required validation.	3	Core data prerequisite.
US-02	Manual cargo entry or small CSV import prototype.	5	Provides initial container records.
US-03	Flag duplicate IDs and missing required fields.	3	Protects data quality.
US-04	Display sample yard blocks and slot states.	5	Makes yard state visible.
US-16	Basic dashboard with vessel count, yard occupancy and stale-data indicator.	5	Demonstrates shared view.

Total planned estimate: 26 story points. This is a sample capacity, not a universal two-week velocity. The team should commit only after reviewing availability and technical uncertainty.

14. SPRINT BACKLOG: TASK SEQUENCE AND DEPENDENCIES

14.1 Suggested two-week sequence

Period	Focus	Expected result
Days 1–2	Sprint planning, field definitions, wireframes, test cases and repository setup.	Stories understood; tasks ready.
Days 3–5	Authentication, vessel-call form and persistence.	Authorized users can create/list vessel calls.
Days 6–7	Cargo entry/import and validation.	Errors are visible and recoverable.
Days 8–9	Yard grid and dashboard summary.	Sample yard and vessel status displayed.
Day 10	Integration testing, bug fixing, review and retrospective.	Demonstrable increment and next-sprint actions.

14.2 Dependencies and risk controls

- Role checks must be in place before protected data-entry features are considered complete.
- The yard dashboard can use clearly labeled simulation data while live interfaces are unavailable.
- Do not present stale or missing data as current; show timestamps and unknown states.
- If CSV import threatens sprint completion, deliver manual cargo entry first and move import refinements to the next sprint.
- Keep advanced optimization out of Sprint 1 until input schemas and constraints are understood.
- Use small, frequent integrations rather than combining all components at the end.

14.3 Daily Scrum questions

- What work moved closer to the Sprint Goal yesterday?
- What will I work on today?
- What blocker or dependency needs help?

14.4 Sprint review and retrospective

At the Sprint Review, demonstrate completed functionality using a realistic but non-sensitive sample scenario and gather stakeholder feedback. In the retrospective, discuss what helped delivery, what caused

delay and one or two specific improvements for the next sprint. The review is about the product increment; the retrospective is about how the team works.

15. STORY-POINT ESTIMATION

Story points estimate relative effort, complexity, uncertainty and risk; they are not hours. This report uses a Fibonacci-like scale: 1, 2, 3, 5, 8, 13. A team can use Planning Poker: each member privately selects a value, the team discusses differences and then agrees on a shared estimate.

Points	Relative size	Example
1	Very small	Change a label or add a simple validation message.
2	Small	Add a simple field to an existing form.
3	Moderate	Create a basic vessel-call form with validation.
5	Medium / multiple concerns	Role-based access or a dashboard using several data fields.
8	Large / uncertain	Constraint-based slot recommendation with multiple cargo rules.
13	Very large / high uncertainty	Broad optimization across cranes, yard and vehicle routes; split before sprint commitment.

15.1 Sprint story estimates

Story	Estimate	Justification
US-19 Role access	5	Authentication integration, role rules and negative authorization tests.
US-01 Vessel call	3	Small data model, form, validation and list view.
US-02 Cargo entry/import	5	Parsing/entry, field mapping and error reporting.

US-03 Cargo validation	3	Duplicate checks and mandatory-field tests.
US-04 Yard grid	5	Slot model, status states and visual representation.
US-16 Basic dashboard	5	Combine counts, status, freshness and access checks.

16. SPRINT GOAL AND EXPECTED DELIVERABLES

16.1 Sprint Goal

By the end of Sprint 1, deliver a secure prototype in which an authorized planner can register a vessel call, enter basic cargo data, see validation errors, inspect a sample yard grid and view a dashboard that clearly distinguishes current sample data from missing or stale data.

16.2 Expected deliverables

- Working prototype login and role checks for agreed roles.
- Vessel-call create/list screens with required-field validation.
- Cargo entry or limited CSV import with row-level error feedback.
- Duplicate and missing-field checks for container records.
- Sample yard grid showing empty, occupied, reserved and unavailable states.
- Dashboard with vessel-call count, yard occupancy summary and data-freshness indicator.
- Audit log for key create/update actions.
- Automated or documented acceptance-test results.
- Sprint Review demonstration, stakeholder feedback and prioritized follow-up backlog.

17. RELEASE ROADMAP AND FUTURE SPRINTS

Increment	Candidate features	Exit evidence
Sprint 1 — Foundation	Access, vessel/cargo entry, validation, sample yard grid, basic dashboard.	Accepted prototype and tests.
Sprint 2 — Yard planning	Slot restrictions, unavailable areas, basic compatible-slot recommendations.	Constraint tests and planner feedback.

Sprint 3 — Crane queue	Priorities, job sequencing, equipment availability and conflict detection.	No accepted schedule contains configured critical conflicts in test scenarios.
Sprint 4 — Vehicle routing	Network graph, blocked lanes, route alternatives and route explanations.	Routes are connected and exclude known blocked segments.
Sprint 5 — Disruption response	ETA change, equipment unavailability, affected-job view and re-planning.	Replanning test results and audit history.
Sprint 6 — Evaluation and pilot readiness	KPI reporting, performance tests, security review and user training.	Pilot-readiness checklist and approved test plan.

17.1 Release strategy

Use staged releases. Begin with a simulation environment and synthetic or approved anonymized data. Compare recommendations with current planning decisions. Conduct user acceptance testing with port staff, resolve safety and data-quality issues, and obtain formal approval before any production integration. Provide a rollback path and clear ownership for incidents.

17.2 Change management

- Explain how recommendations are generated and when human approval is required.
- Train users by role and provide concise quick-reference guides.
- Record feedback and convert recurring issues into backlog items.
- Maintain configuration versions for constraints, priority rules and route restrictions.
- Communicate downtime, data-source failures and plan-version changes.

18. RISKS, MITIGATIONS AND ETHICAL CONSIDERATIONS

Risk	Potential impact	Mitigation
Incorrect or stale inputs	Invalid slot or schedule recommendation.	Timestamp data, validate fields, flag stale state and show unknowns.

Unsafe schedule or route	Equipment conflict or hazard exposure.	Hard constraints, independent validation, human approval and simulation tests.
Optimization too slow	Plan is outdated before use.	Start with heuristics, define response targets and return a feasible fallback when appropriate.
Integration failure	Dashboard becomes incomplete.	Adapters, retries, health status and clear degraded-mode behavior.

18.1 Ethical and operational safeguards

- The platform should support qualified staff rather than obscure accountability.
- Every important recommendation should expose key reasons and constraint failures.
- Access to cargo, vessel and user information should follow approved policy.
- The system should not hide uncertainty or invent missing sensor values.
- Safety-critical controls remain subject to port standards, equipment vendor requirements and formal engineering revie

19. TESTING AND VALIDATION PLAN

Test level	Example	Pass condition
Unit tests	Validate required fields and duplicate container IDs.	Expected error or success result for every case.
Integration tests	Cargo import updates the shared cargo list and dashboard.	Data is consistent across modules.
Acceptance tests	Planner registers vessel and sees yard/dashboard state.	Story acceptance criteria pass.
Constraint tests	Try overlapping crane jobs or assign a blocked lane.	Invalid plan is flagged and not approved.
Performance tests	Load a defined sample number of jobs and dashboard events.	Response time meets the agreed test target.

Security tests	Attempt unauthorized access and invalid sessions.	Access denied and event logged as appropriate.
Recovery tests	Simulate data-feed loss or delayed updates.	Stale state is visible; service recovers without silent corruption.
Usability tests	Ask representative users to complete common planning tasks.	Tasks completed and feedback recorded.

20. CONCLUSION

The Seaport Container Yard Coordination and Optimization Platform addresses a coordination problem across vessel planning, cargo records, yard storage, cranes, terminal vehicles and operational dashboards. Its value comes from sharing reliable state, detecting constraints early, proposing coordinated actions and making changes traceable.

The Agile approach converts the broad project into epics, user stories and measurable acceptance criteria. The prioritized backlog puts security, vessel/cargo data and operational visibility ahead of complex optimization. The first sprint creates a secure foundation that can be demonstrated and reviewed by port stakeholders. Later sprints can add slot recommendations, crane scheduling, vehicle routing and disruption recovery in controlled increments.

22. REFERENCES

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